

Technology Overview: *Centralized Computing Platforms*

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DELPHI

Innovation for the Real World

Forward-looking statements

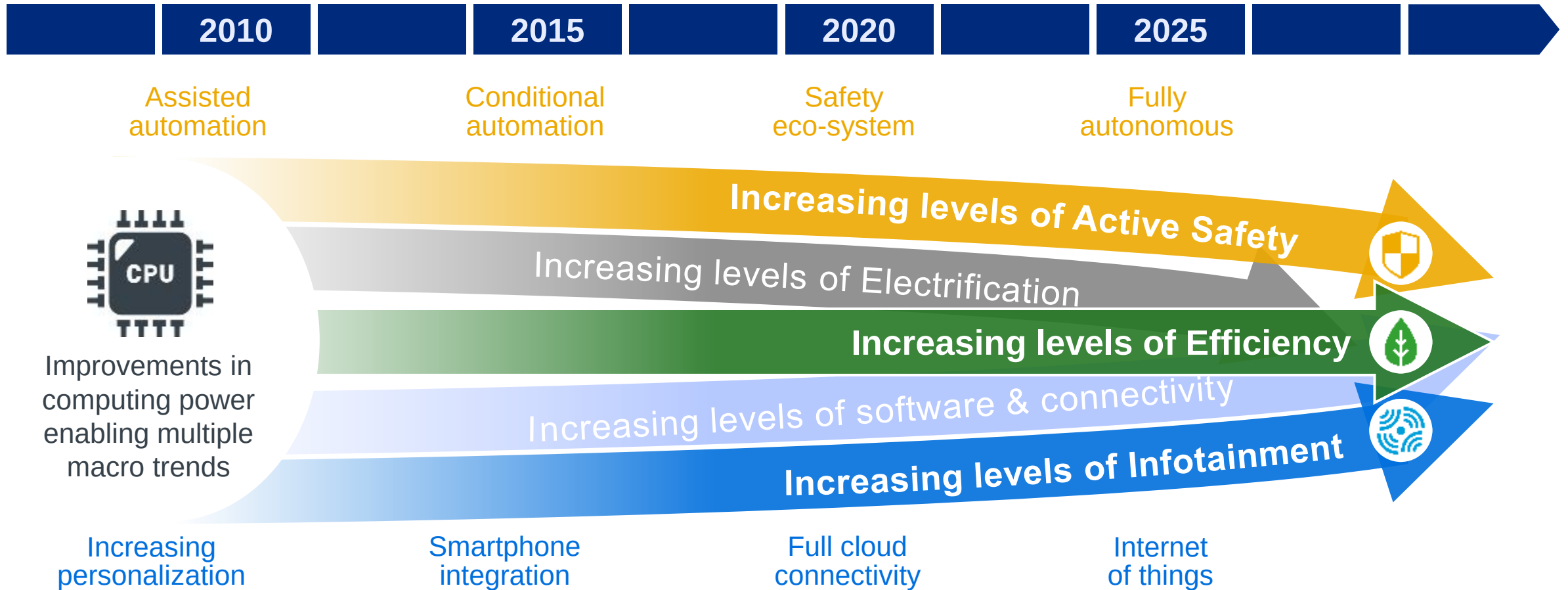
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Overview

- Vehicle automation increasing computing and electrical architecture demands
 - Vehicles increasingly “smart”; massive amount of computing power going into the vehicle
 - Systems integration capabilities critical to optimize performance, mass and cost
 - Delphi well positioned with foundation in computing, signal and power distribution
- Accelerating demand for “smartphone like” experience, connectivity and consolidation
 - Seamless integration into the vehicle environment remains a differentiator
 - New more powerful compute platforms enabling the next level of integration
 - Delphi V2X tech enabling connectivity between autos and infrastructure; creating new challenges and opportunities
- Powertrain electrification required to enable efficiency and control
 - “Drive by wire” and sensor power demands increasing electrical demands
 - Emissions and fuel efficiency regulations driving demand for cost effective green solutions
 - Delphi comprehensive Powertrain components portfolio enabling intelligent electrification

Convergence of safe, green and connected

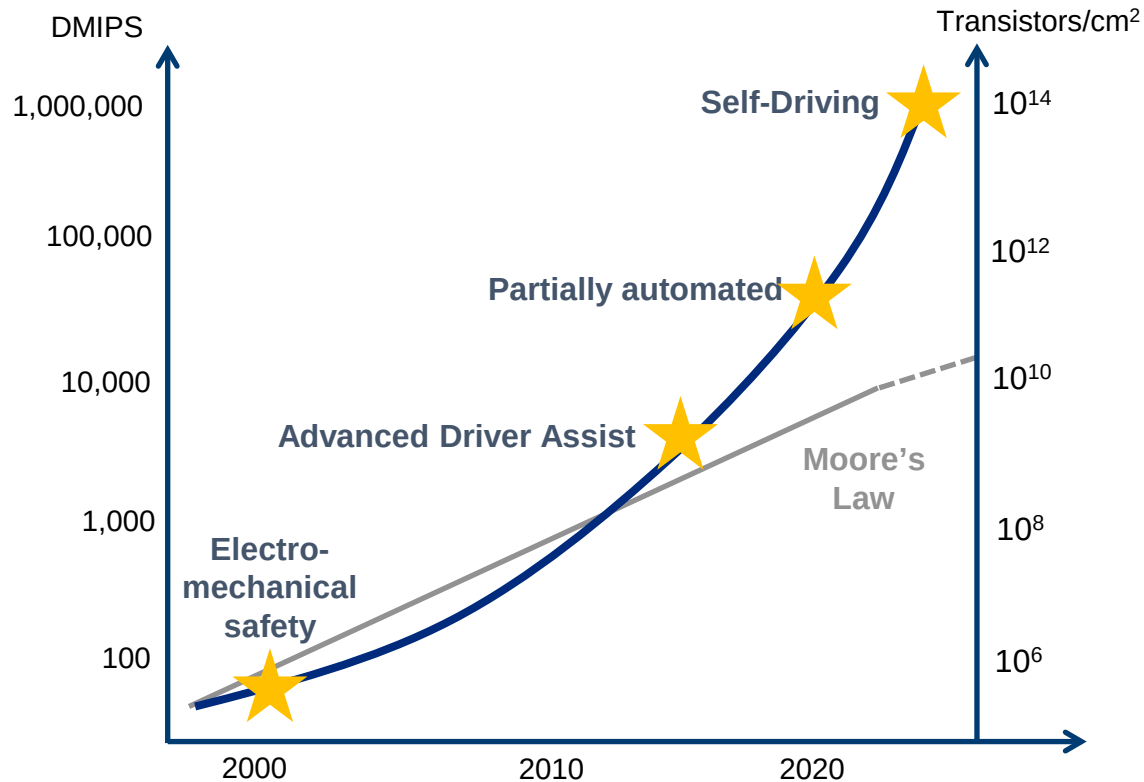
Megatrends driving computing power requirements



Convergence of megatrends driving demand for increased computing power

Exponential computing power

Computing power



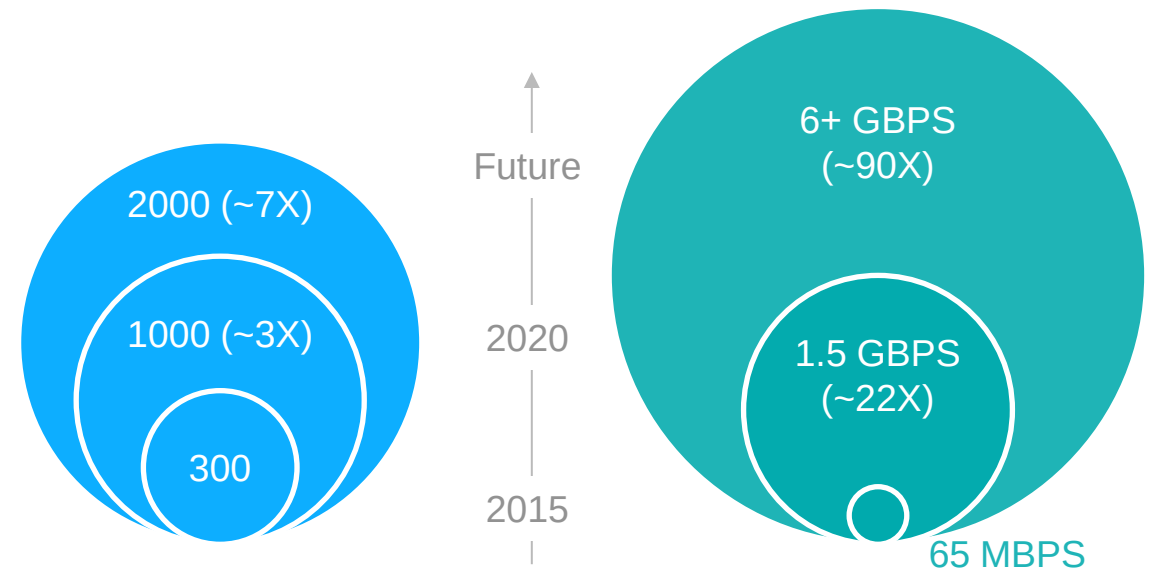
Architecture capabilities

Diagnostic parameters

Rapid increase in number of systems monitored helps provide early warnings of issues

Data transfer speeds

High speed Ethernet replaces antiquated 1980s technology

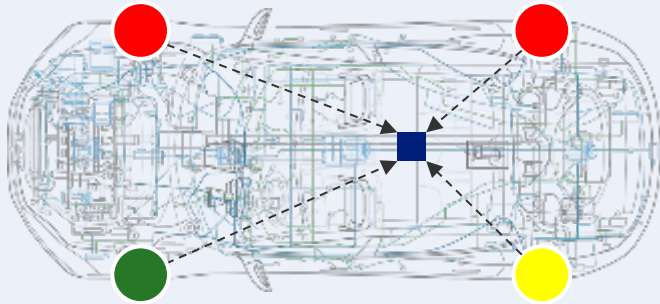


Rapid acceleration in automotive computing power

Centralized computing demand drivers

Managing complexity

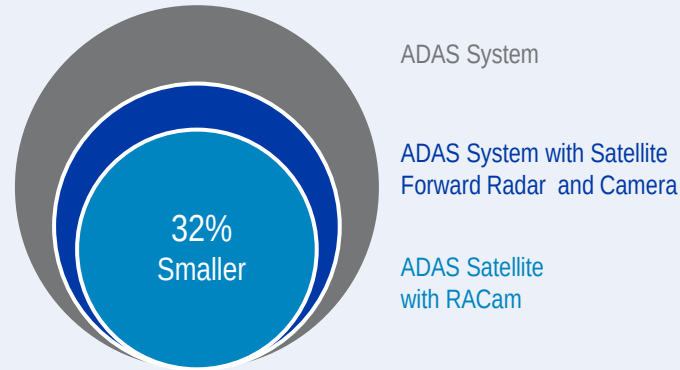
Simplifying architecture; managing data speeds, complexity



- Centralized decision making provides multiple data-points in decision making
- Strategically locating computers helps minimize cable length, unlock content growth

Lower mass

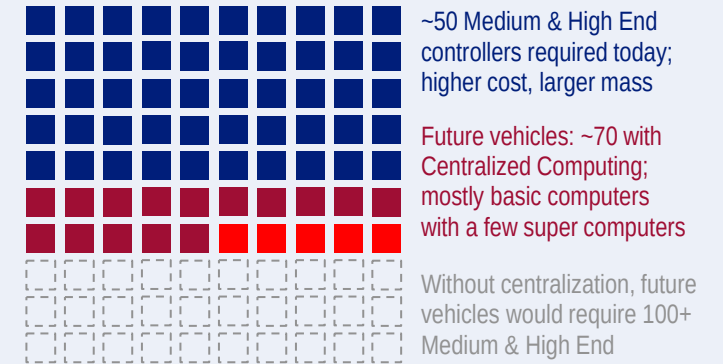
"Basic" sensors and optimized architecture



- Centralizes computing power onto chip
- Sensors become more basic (smaller & lighter) and easier to integrate

Reduced cost

Meet future computing demands with fewer controllers

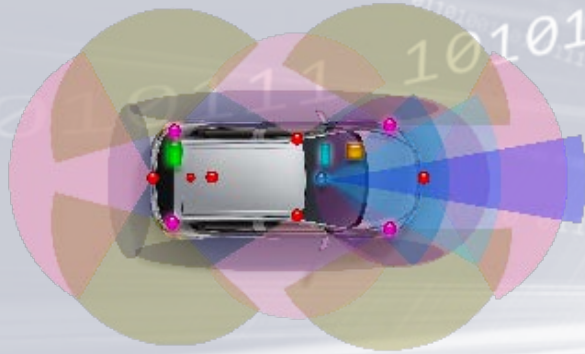


- Reduces incremental controllers needed
- Focuses workload on handful of supercomputers

~30% savings from lower mass and cost

Delivering technology innovation

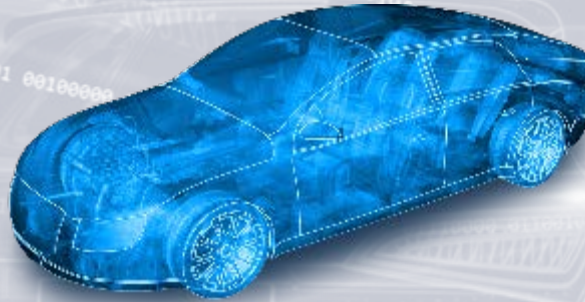
Automated capabilities



Sensor data fusion enabling:

- Blind spot warning
- Adaptive cruise control
- Lane departure warning / prevention
- Forward collision warning
- Automatic emergency braking
- Collision mitigation

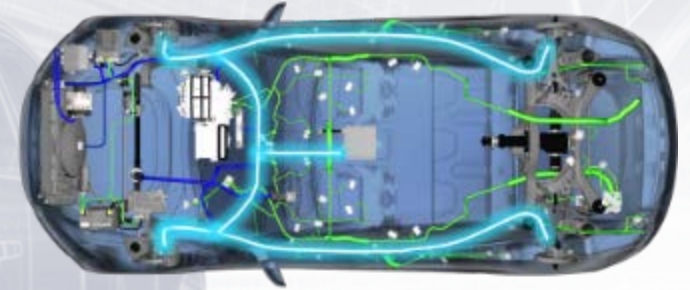
Infotainment



Cockpit integration:

- V2X Intelligent Driving
- Eye glance
- Gesture control
- Voice recognition
- On Board Diagnostics
- Customized / 3D displays

Intelligent electrification



Sophisticated software driving:

- Engine control
- Valve Train and engine sensors
- Fuel injection systems
- DC/DC converter
- Connected Infotainment
- Active Safety

Innovation creates value through best-in-class technologies

Computing platform: Multi Domain Controller (MDC)

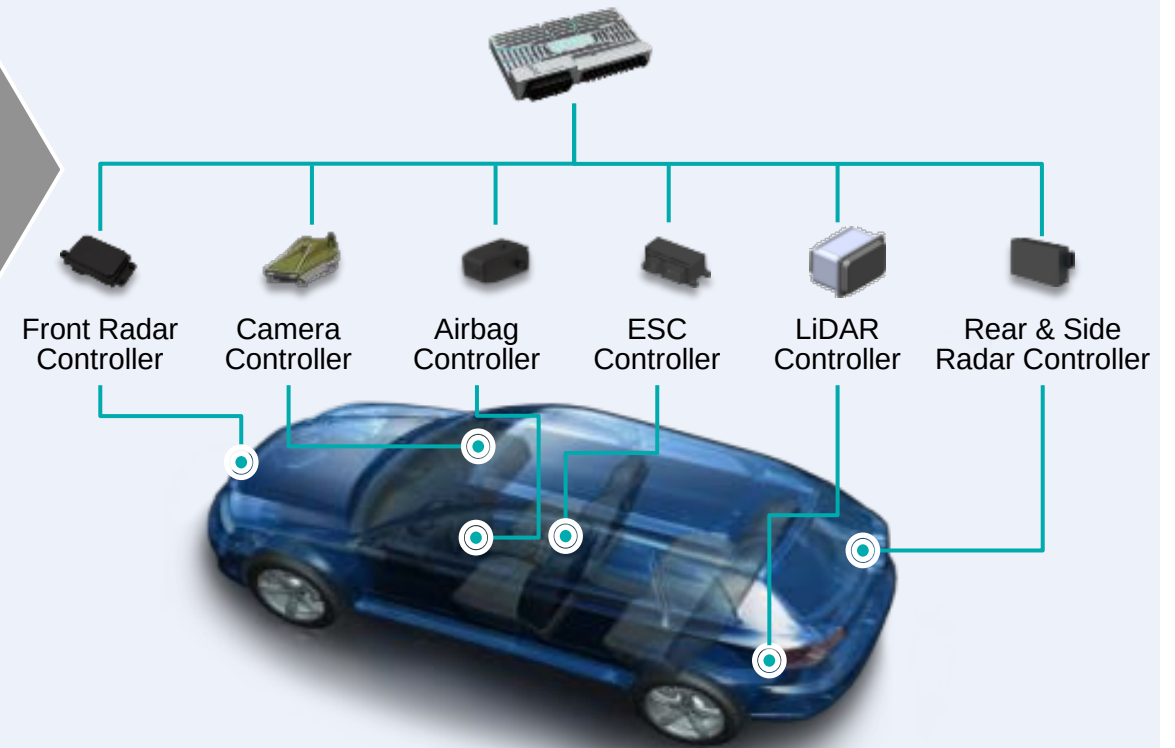
Active Safety Multi-Domain Controller (MDC)

Production
2017
Launch



- Scalable software platform
- Reduced architecture complexity
- Faster communication/interconnection
- Multi-processor configurations

ADAS Centralized Sensor Fusion/Control



MDC reduces complexity and enables future expandability

CSLP platform and partnerships

Delphi

- Radar & Lidar Systems & Fusion
- Route Planning and Behavioral Policy
- Vehicle control and Interface

Mobileye

- Vision Systems & Fusion
- Reference Learning / AI
- Localization using REM

Intel

- Automated Driving SoC
- Connectivity
- Data Centers / Management

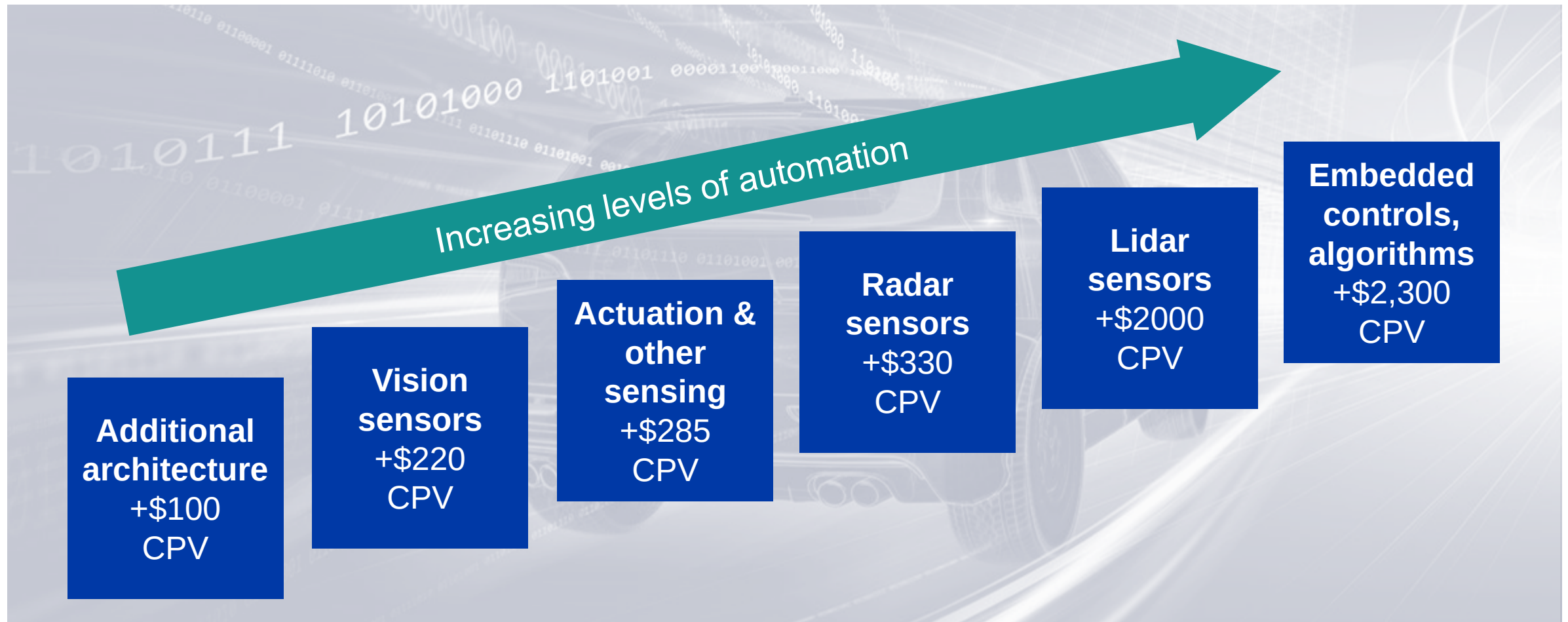
Intel / Delphi / Partners

- OS and support Software
- Middleware and Tools
- Functional Safety



Key strategic partnerships enabling the CSLP platform execution

Increasing content per vehicle – Automated Driving



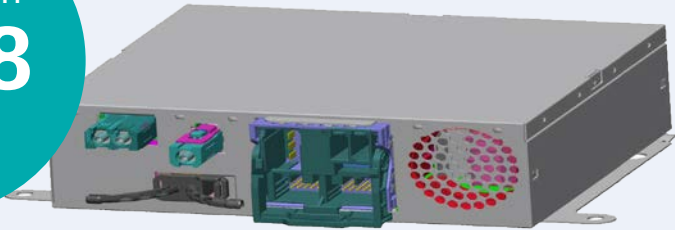
~\$5,000 CPV when moving to a Level 4/5 vehicle in 2020

Note: assumes 2020 volumes; and four lidar sensors, five radar sensors and six vision sensors

Computing platform: Integrated Cockpit Controller (ICC)

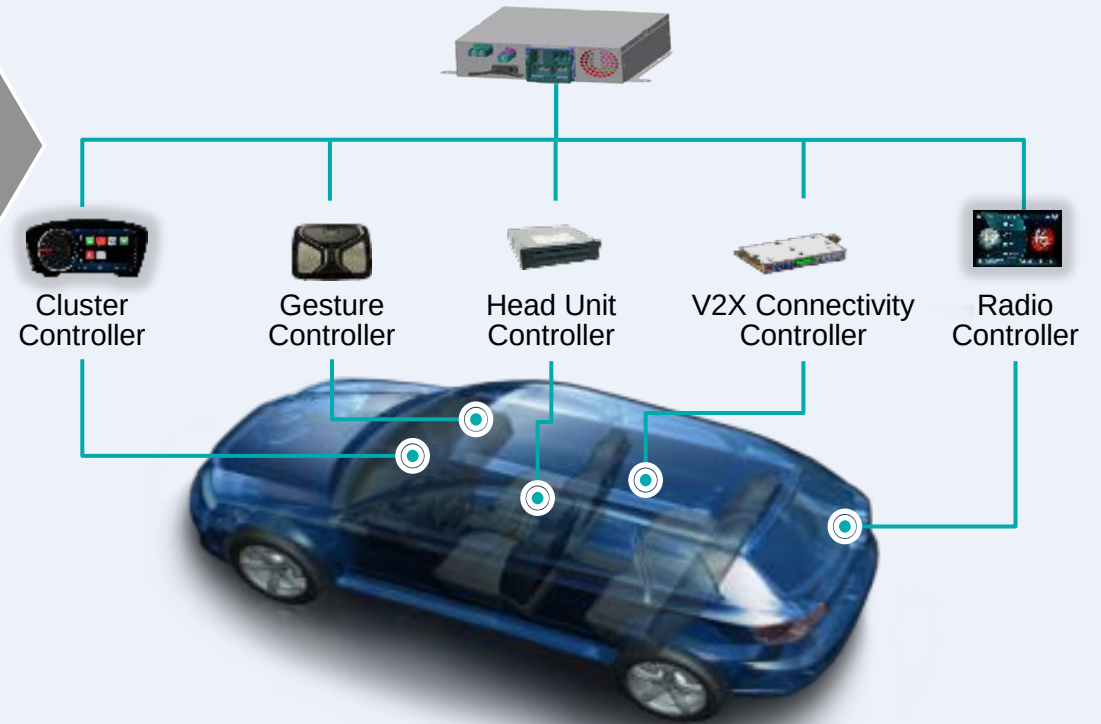
Integrated Cockpit Controller (ICC)

Production
2018
Launch



- Scalable software platform
- Reduced architecture complexity
- Faster communication/interconnection
- Multi-processor configurations

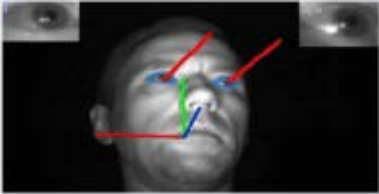
Centralized sensor fusion & user experience control



Future system software optimization and upgradeability

Infotainment computing platform and software architecture

Eye Glance



Reconfigurable / 3D Cluster



Gesture Control



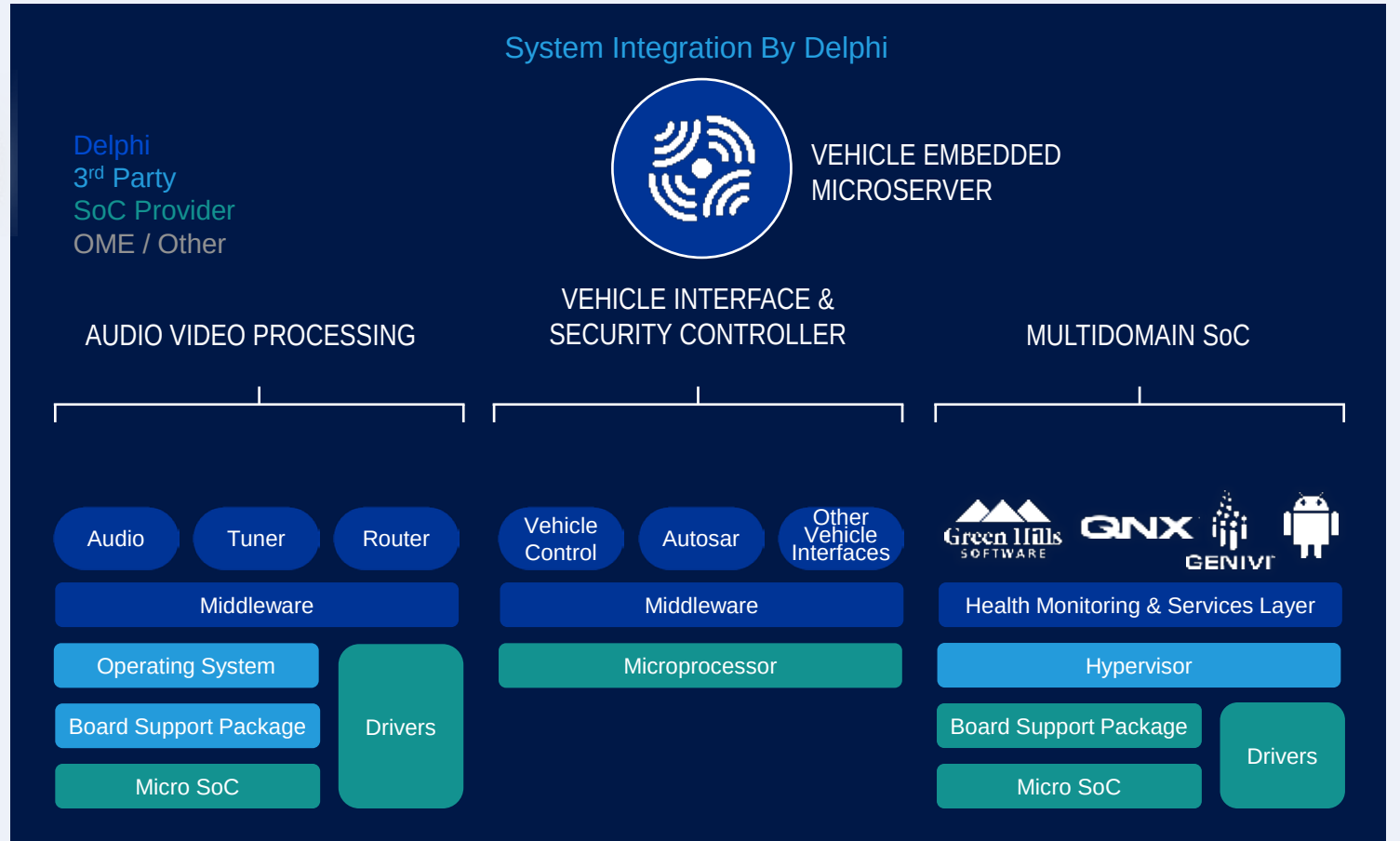
Integrated Center Panel



Voice Recognition

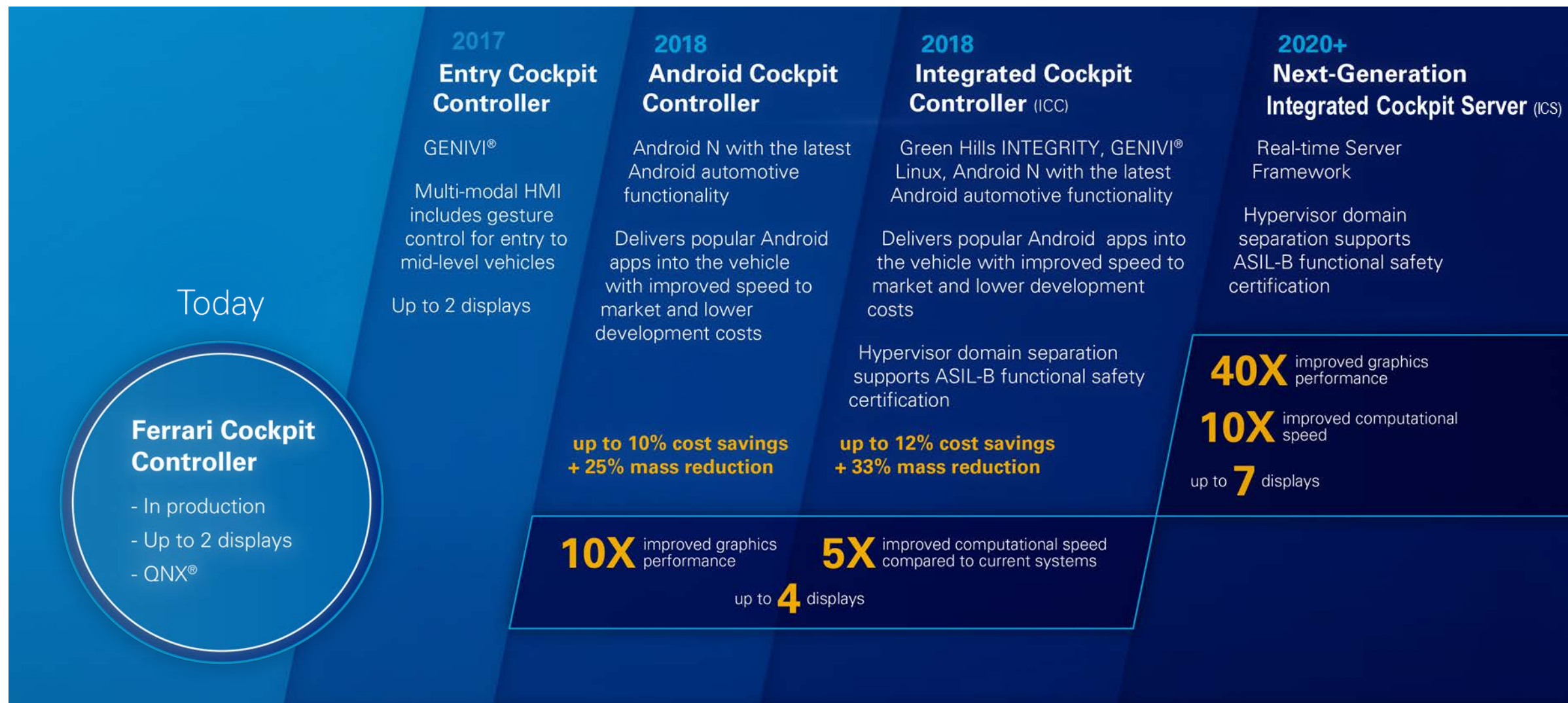


Cloud Connectivity



Goal: Become the industry standard for infotainment platforms

Delphi's approach to Cockpit Architecture



Powertrain Domain Controller (PDC)

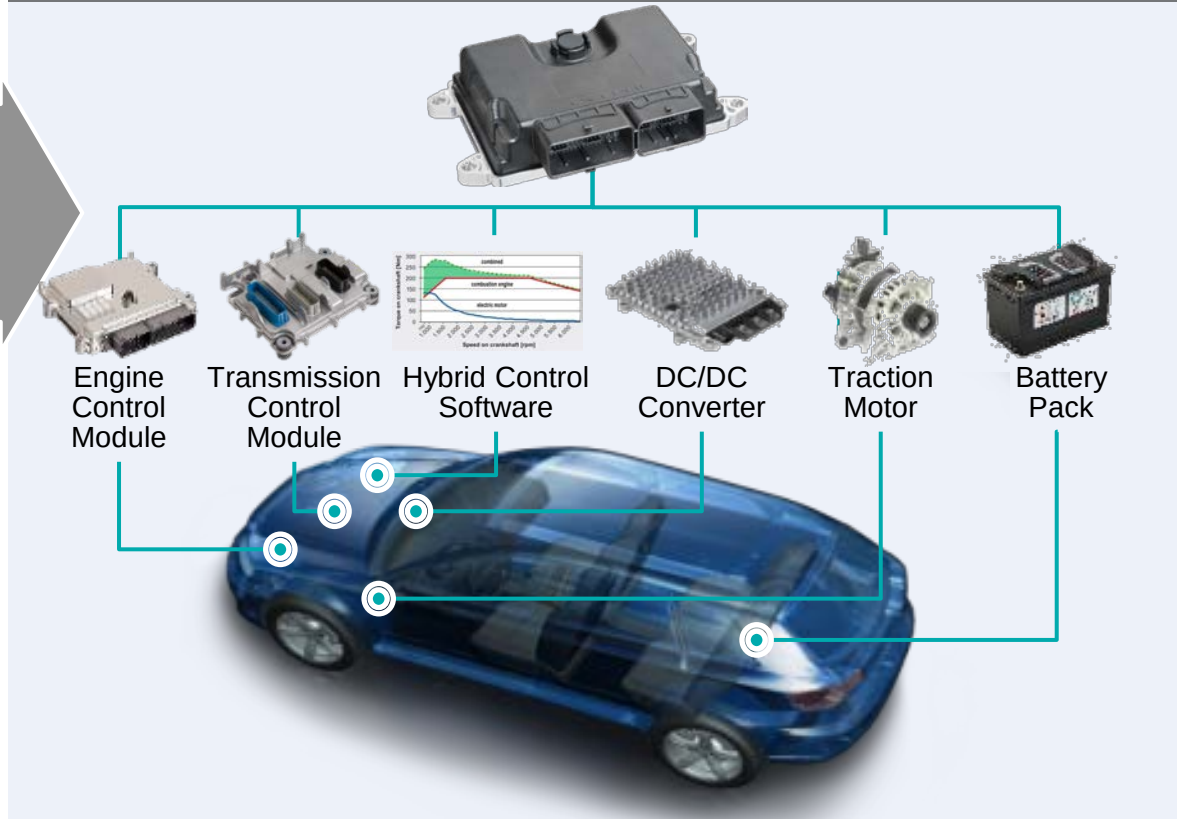
Powertrain Domain Controller (PDC)

Production
2017
Launch



- Powertrain domain control for HEV, PHEV, and EV vehicles
- Consistent, centralized platform to manage complexity
- Integrated interface between Powertrain and Vehicle in EV's
- Delivers torque required with optimal engine and electrification design

Centralized powertrain control



Centralizing powertrain control and optimization

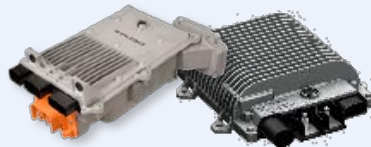
Creating intelligent electrification with Delphi's 48V solution

Enabling intelligent electrification

Engine Management Systems



Converters & Inverters



Electrical Architecture



Connected Infotainment



Valve Train & engine sensors



Active Safety



Enabling power intensive features



CONSUMER PREFERENCE

Active Safety

Infotainment

Performance

Cost effective efficiency



MEETING REGULATIONS

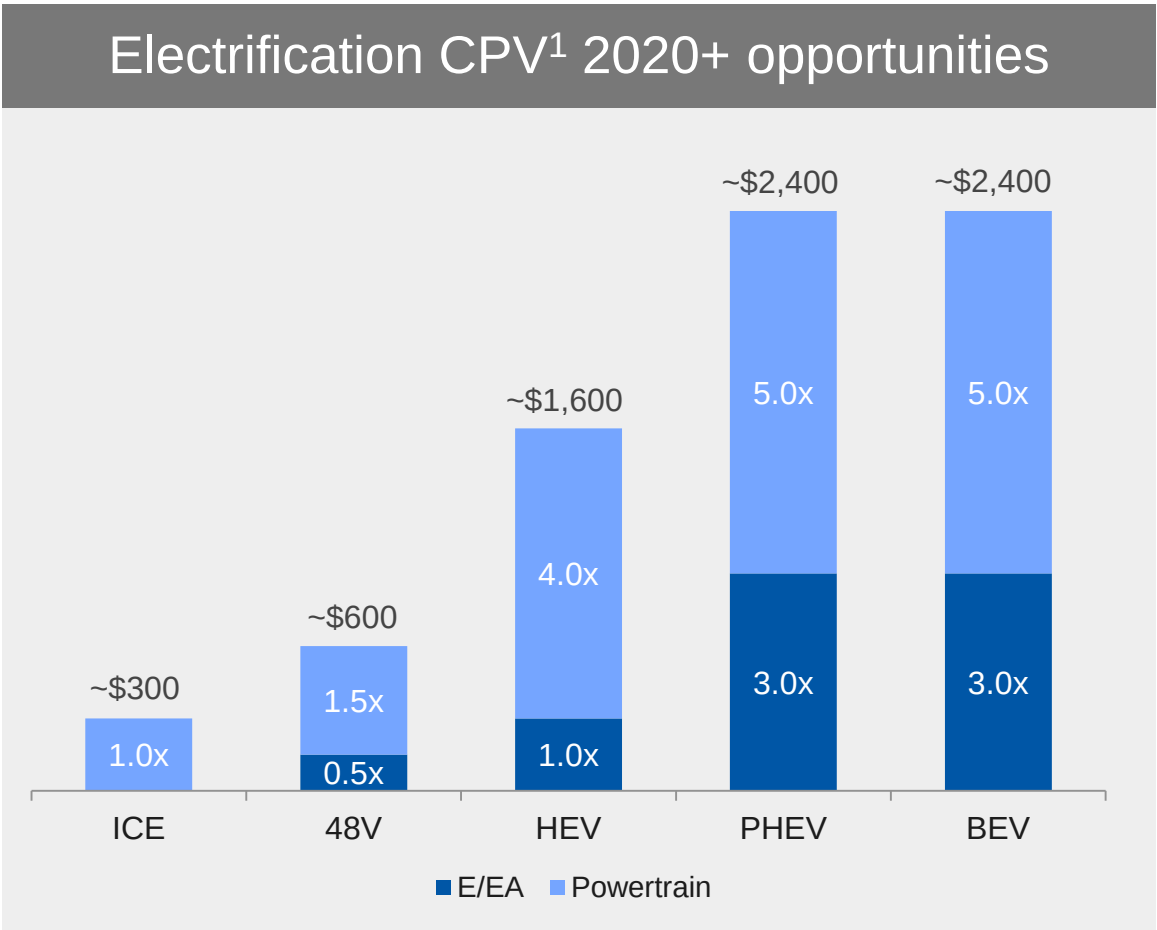
Emissions

Fuel Economy

Software and hardware optimized electrification architecture

Electrification increases Delphi content per vehicle (CPV)

Delphi technologies	
Powertrain Management	
- Supervisory controller - Supervisory software - Inverter - Battery pack controller	- Combined inverter/converter 48V Driver module - DC/DC converter - On-board charger
Electrical/Electronic Architecture	
48V fusing & distribution - High power & voltage connectors - Charging inlets and cables	- High voltage shielded cable - Internal battery connections 12V battery monitor



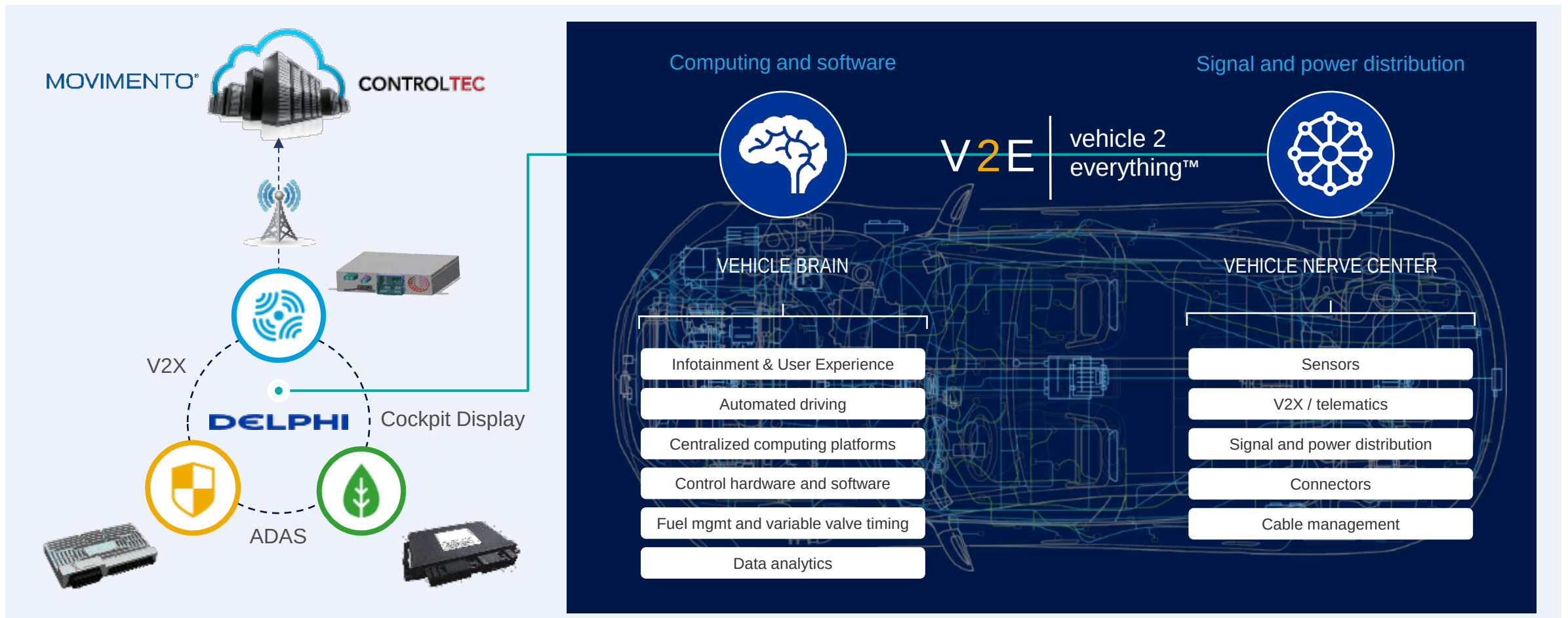
Comprehensive portfolio enables vehicle electrification

1 Content per vehicle multiples represent Total Addressable Market (TAM) for electrified vehicles
Note: Content per vehicle multiples calculated off ~\$300 content on a base gasoline GD_i, 2-step variable valvetrain internal combustion engine in 2023 and beyond

Power of data analytics

CONTROLTEC		DELPHI	MOVIMENTO [®] Ready. Connect. Go.™
• Leverages real-time data and the cloud to identify and solve issues • Broadens our “Big Data” capabilities		• Scalable vehicle software enhances Services business revenues via expanding usage with global OEMs • Continued organic and inorganic investment in software and service	• Provider of telematics, analytics and OTA update technologies • Opportunity to drive market adoption
Enabling capabilities	Over-the-air updates	• Manage update/upgrade campaigns for OEMs • Deliver and validate software updates/upgrades on vehicles	
	Cybersecurity	• Detect and defeat unauthorized intrusions into the vehicle • Combine hardware and software capabilities to develop secure platform	
Commercial opportunities	Fleet management	• Support fleet compliance with regulations for CVs • Enable total-cost-of-ownership savings by improving driver behavior, routes/planning	
	Data brokerage	• Leverage vehicle data to deliver value to third parties (insurance, etc.)	
Great opportunities with more to come			

Delphi vision



Computing capabilities enabling intelligent mobility solutions

Computing power enabling Safe, Green and Connected

Convergence of megatrends driving exponential computing power demand

Reducing electrical architecture complexity is a “must do” to enable content growth

Delphi Multi Domain Controller unlocking ADAS capabilities

Delphi Integrated Cockpit Controller integrating Infotainment features

Delphi Power Domain Controller enabling intelligent electrification

Delphi competitive advantages critical for future mobility

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